Artificial Intelligence

Beyond Classical Search

Genetic algorithms

- A genetic algorithm is a variant of stochastic beam search in which successor states are generated by combining two parent states rather than by modifying a single state.
- A successor state is generated by combining two parent states
- Start with *k* randomly generated states (population)
- A state is represented as a string over a finite alphabet (often a string of 0s and 1s)
- Evaluation function (fitness function). Higher values for better states.
- Produce the next generation of states by selection, *crossover*, and *mutation* (*genetic operators*)

Biology Concepts

- Population
- Fitness
- Selection
- Crossover
- Mutation

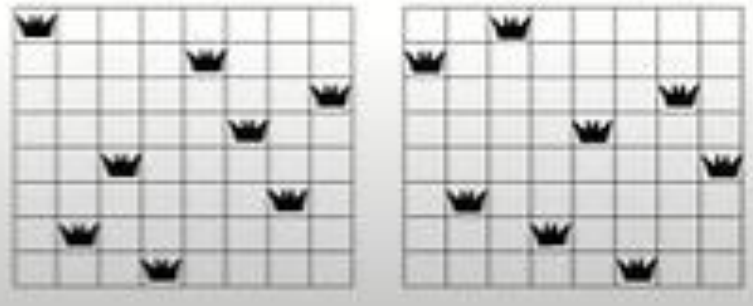
Biology Concepts: Population

Biology

- Collection of individuals.

Algorithm

- Collection of states.



Biology Concepts: Fitness

Biology

- More healthy, less prone to diseases.

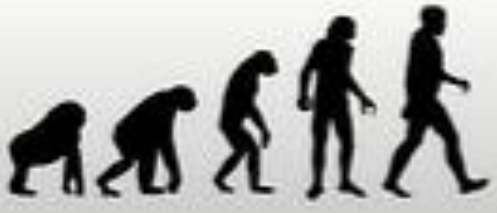
Algorithm

- Closest to the final solution.

Biology Concepts: Selection

Biology

- Selecting species that are the most biologically fit.



Algorithm

- Selecting states that are closest to the solution (Fittest).

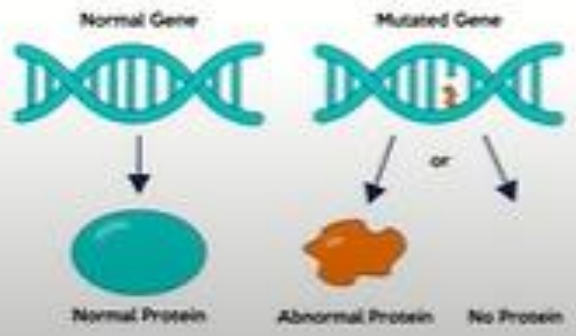
Biology Concepts: Crossover

Biology	Algorithm
generation	Interchanging values between selected states

Biology Concepts: Mutation

Biology

- Change or variation.



Algorithm

- Alteration.

Before Mutation

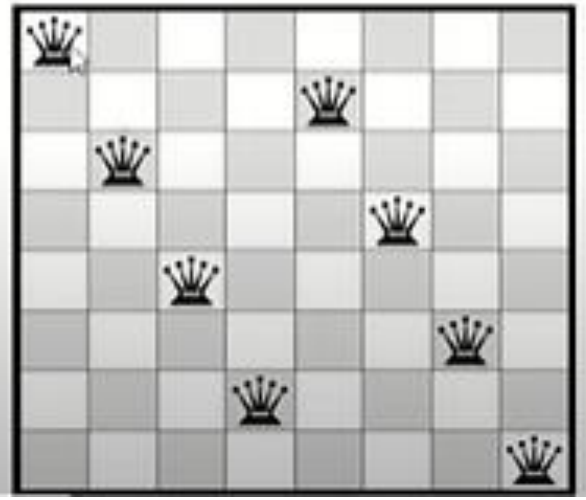
A5	1	1	1	0	0	0

After Mutation

A5	1	1	0	1	1	0

8- Queens Problem

- Arrange 8 queens on a standard chess board in such a way that no queen attacks each other.



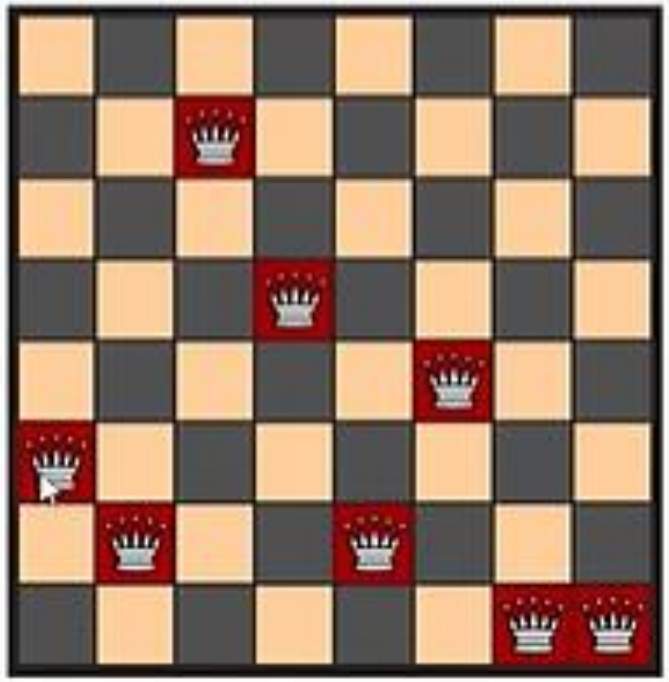
Solving 8-Queens using Genetic Algorithm

- Step 1: Representing individuals.
- Step 2: Generating an initial Population.
- Step 3: Applying a Fitness Function.
- Step 4: Selecting parents for mating in accordance to their fitness.
- Step 5: Crossover of parents to produce new generation.
- Step 6: Mutation of new generation to bring diversity.
- Step 7: Repeat until solution is reached.

Step 1: Representing Individuals/States

- Formulate an appropriate method to represent individuals of a population.
- Array.
- Index: Column.
- Value: Row.

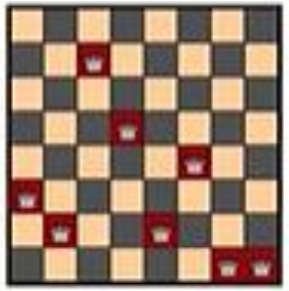
3	2	7	5	2	4	1	1



Step 2: Generate Initial Population

- Generate random arrangements of 8 queens on a standard chess board.

A



3	2	7	5	2	4	1	1

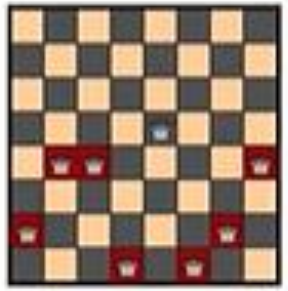
B

2	4	7	4	8	5	5	2

C

3	2	5	4	3	2	1	3

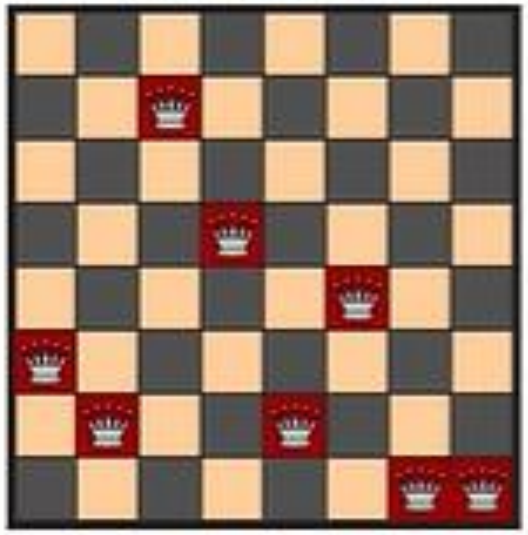
D



2	4	4	1	5	1	2	4

Step 3: Apply Fitness Function

Individual



3	2	7	5	2	4	1	1

Fitness = No. of non attacking pairs

- **Queen 1:** 6
- **Queen 2:** 5
- **Queen 3:** 4
- **Queen 4:** 3
- **Queen 5:** 3
- **Queen 6:** 2
- **Queen 7:** 0
- **Queen 8:** 0

Total 23

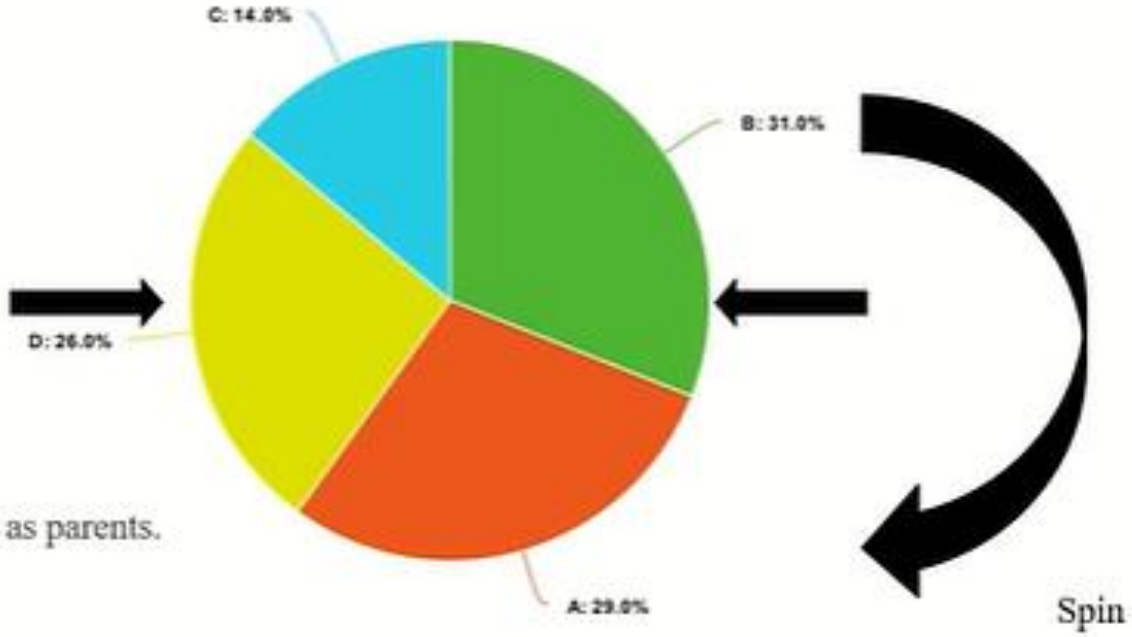
Step 3: Apply Fitness Function (contd.)

	A	B	C	D
Individuals				
	3 2 7 5 2 4 1 1	2 4 7 4 8 5 5 2	3 2 5 4 3 2 1 3	2 4 4 1 5 1 2 4
Fitness	23	24	11	20
Fitness %	29%	31%	14%	26%

Total:78

Step 4: Selection

Proportionate selection or roulette wheel selection gives individuals **with higher fitness values** a higher chance of being selected as parents, mimicking the concept of "survival of the fittest" in the natural evolutionary process.

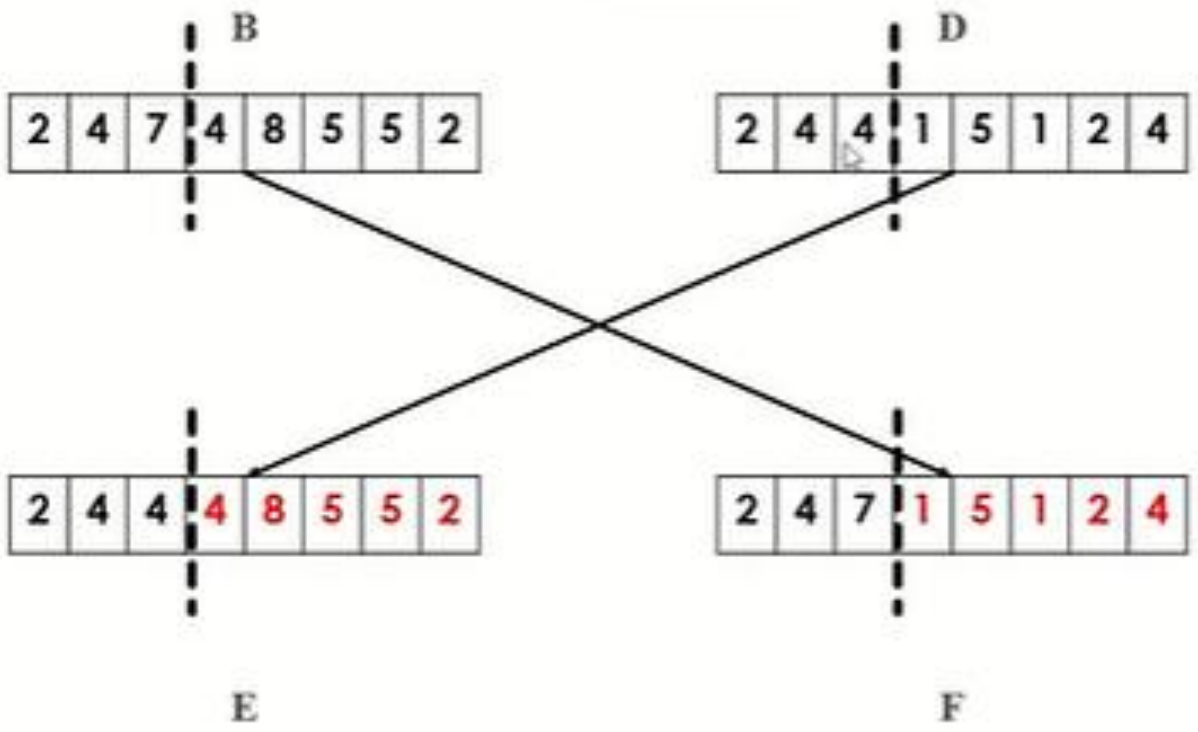


- B and D are selected as parents.

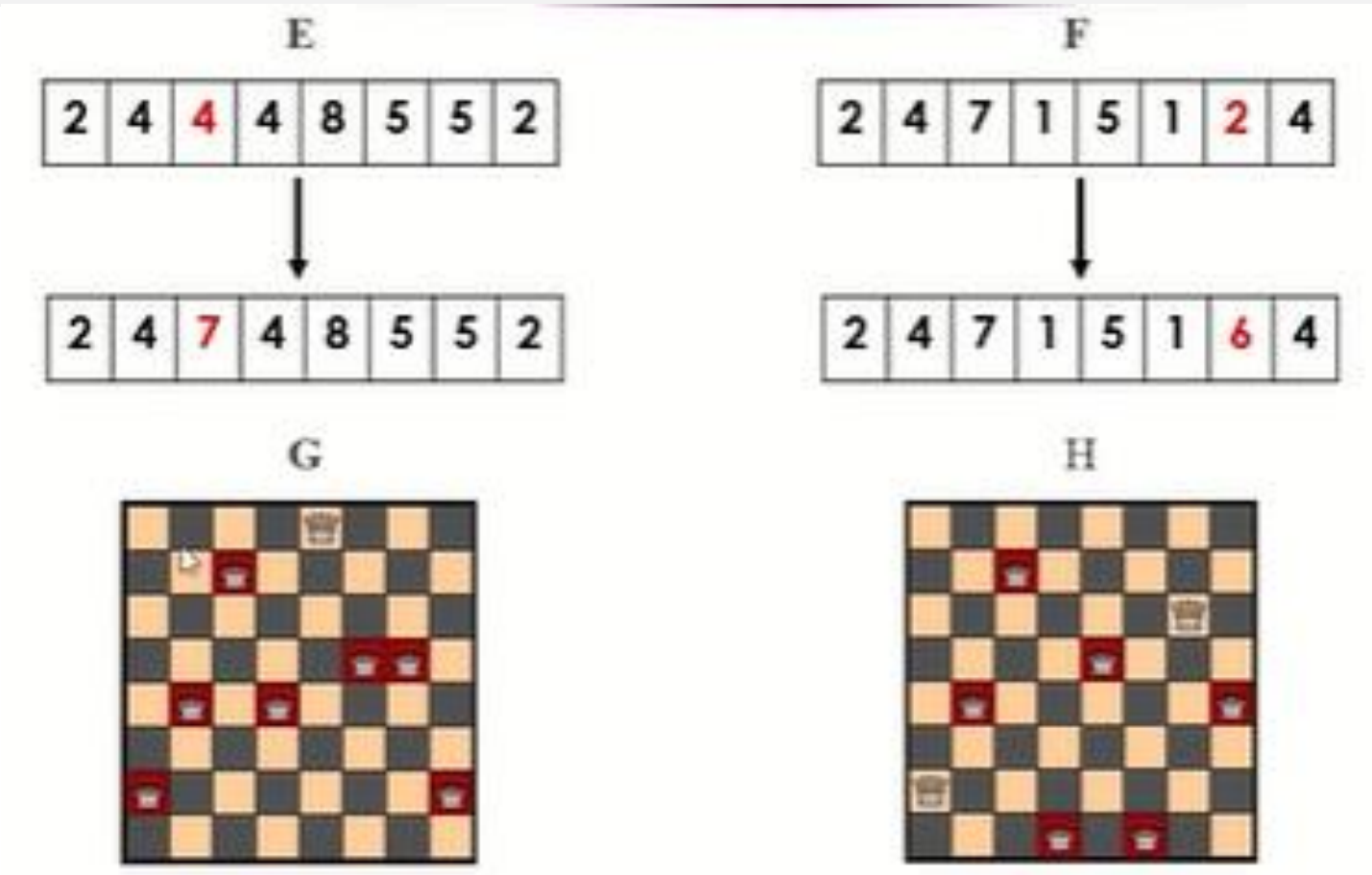
- There are various methods of selection.
- Roulette Wheel, Tournament, Rank, etc.
- Stochastic Universal Sampling (SUS).
- Population is divided on a wheel according to their respective percentages of fitness and two fixed points are placed.
- Wheel is spun and those individuals are selected at which the fixed points are pointing when the wheel stops.

The key difference between SUS and traditional roulette wheel selection is that **SUS selects multiple parents in a single pass, rather than selecting one parent at a time.**

Step 5: Crossover



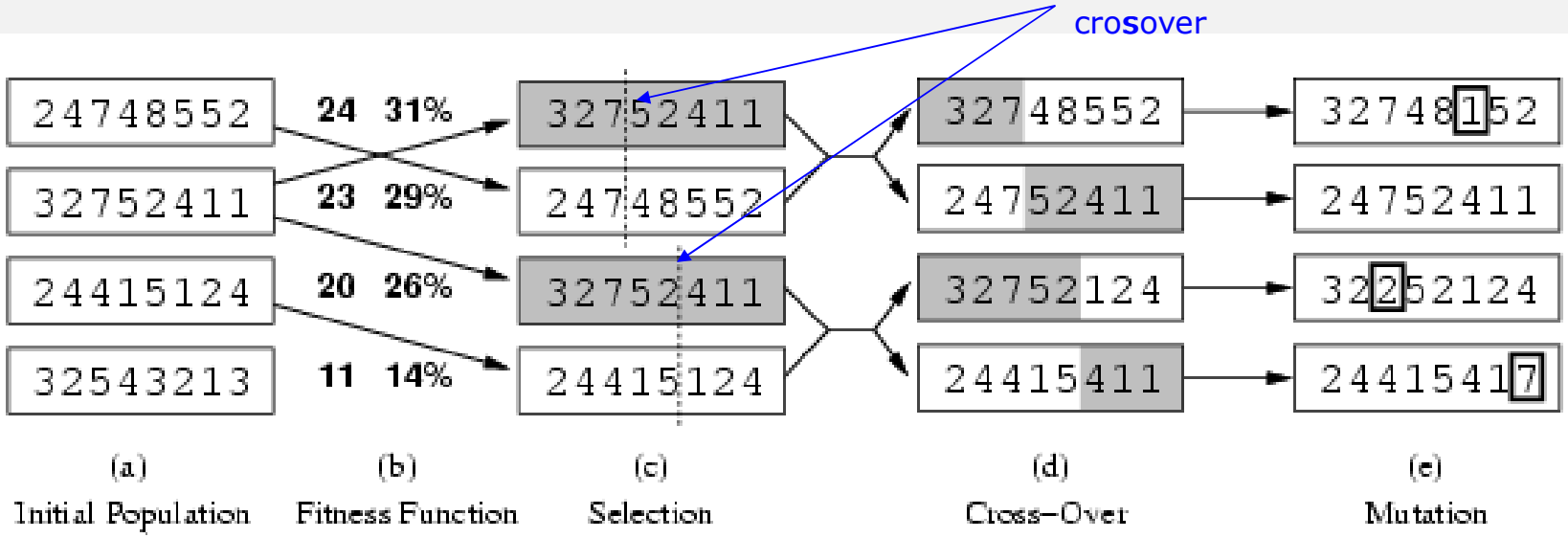
Step 6: Mutation



Step 7: Repeat

- All steps are repeated until best solution is reached.
- Best solution = Highest fitness score (28 in this case).

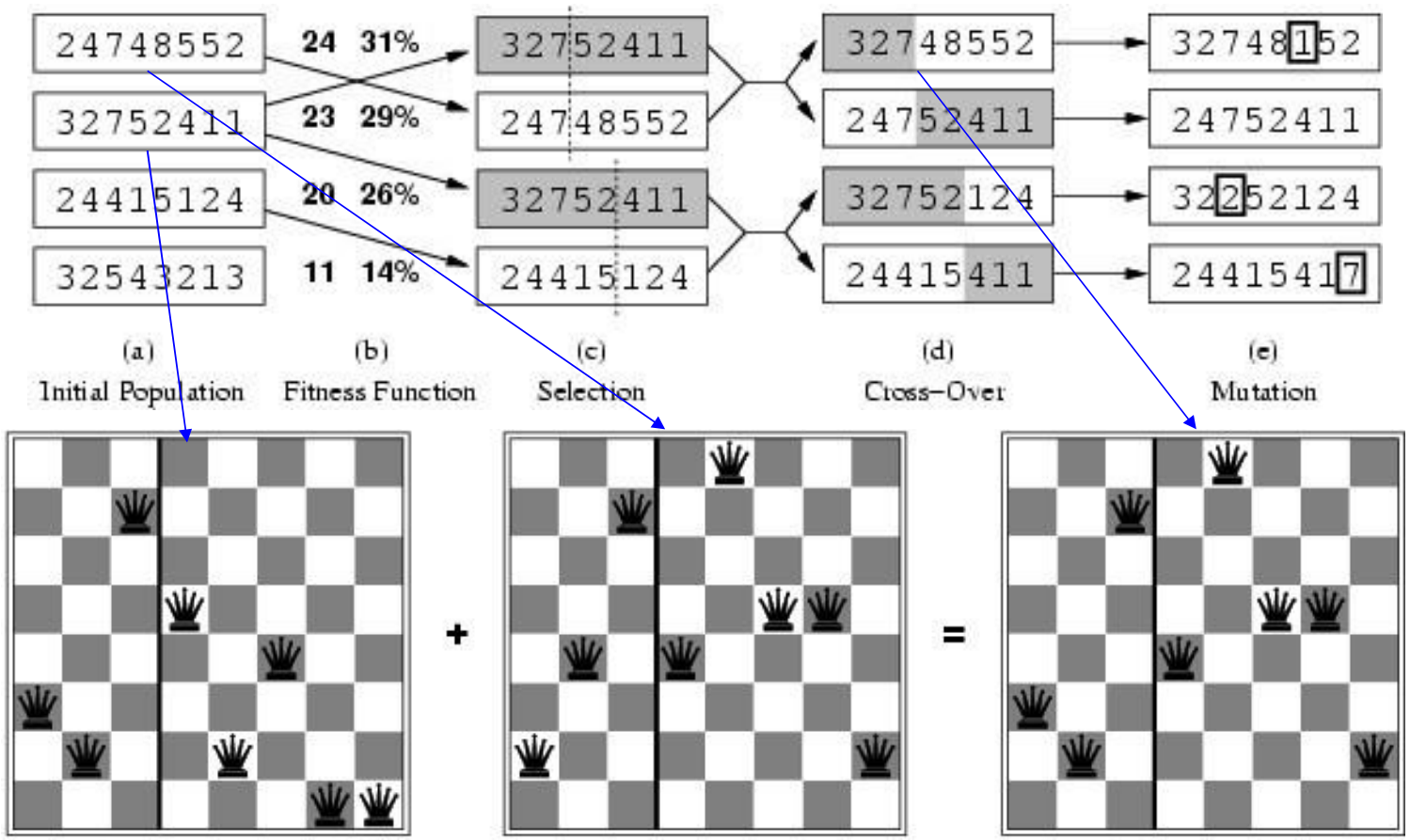
Genetic algorithms



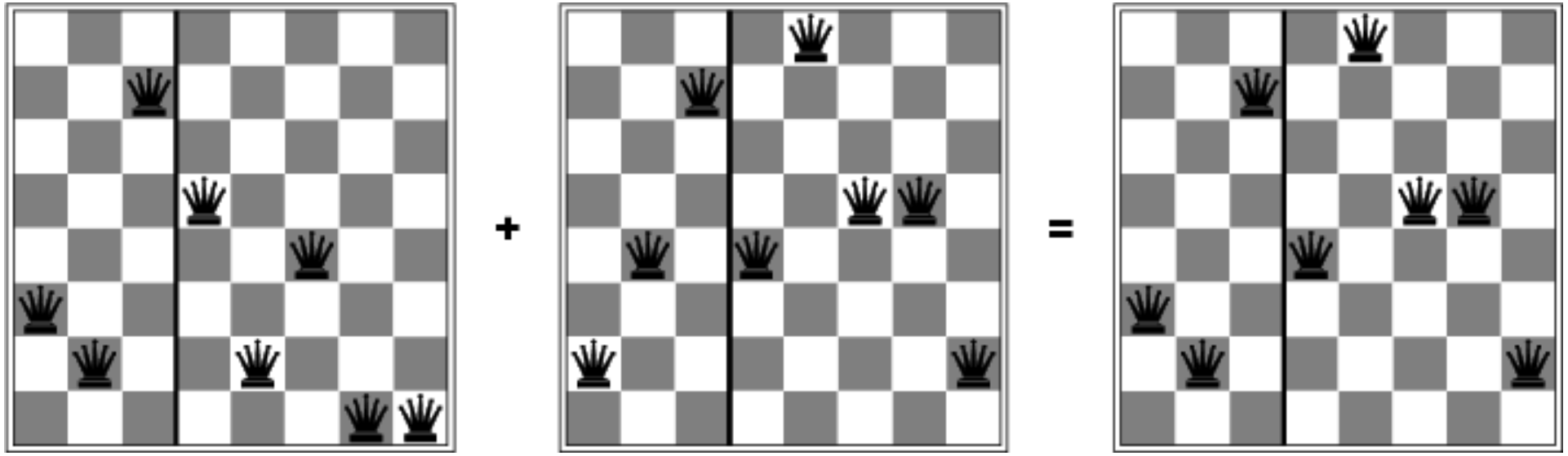
Example 8-Queen

- Fitness function: number of non-attacking pairs of queens (min = 0, max = 28)
- 24/(24+23+20+11) = 31%
- 23/(24+23+20+11) = 29% etc

8-Queen Example



Genetic algorithms



Algorithm

```
function GENETIC-ALGORITHM(population, FITNESS-FN) returns an individual
  inputs: population, a set of individuals
          FITNESS-FN, a function that measures the fitness of an individual

  repeat
     new_population ← empty set
     for i = 1 to SIZE(population) do
        x ← RANDOM-SELECTION(population, FITNESS-FN)
        y ← RANDOM-SELECTION(population, FITNESS-FN)
        child ← REPRODUCE(x, y)
        if (small random probability) then child ← MUTATE(child)
        add child to new_population
     population ← new_population
  until some individual is fit enough, or enough time has elapsed
  return the best individual in population, according to FITNESS-FN
```

```
function REPRODUCE(x, y) returns an individual
  inputs: x, y, parent individuals

  n ← LENGTH(x); c ← random number from 1 to n
  return APPEND(SUBSTRING(x, 1, c), SUBSTRING(y, c + 1, n))
```

Figure 4.8 A genetic algorithm. The algorithm is the same as the one diagrammed in Figure 4.6, with one variation: in this more popular version, each mating of two parents produces only one offspring, not two.

Applications: Genetic algorithms

- DNA Analysis
- Image Processing
- Vehicle Routing
- Neural Network

Solving Difficult Problems

GAs prove to be an efficient tool to provide **usable near-optimal solutions** in a short amount of time.

Failure of Gradient Based Methods

In most real-world situations we have a very complex problem as it suffer from an inherent tendency of getting stuck at the local optima as shown here.

Getting a Good Solution Fast

Some difficult problems like the Travelling Salesperson Problem (TSP), have real-world applications like path finding and VLSI Design.

Homework

- Readings
  - CH 4 Section 4.1